

Week 3 Notes

Mathematics I: Quadratic Functions

Course: Mathematics I — IIT Madras BS Programme

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Lecture 1 — Introduction to Quadratic Functions

Definition, Graphing, Axis of Symmetry, Vertex & Range

1 Recap: Where We Left Off

From Points to Lines to Quadratics

We began with simple geometric objects (points and lines), plotted them on the coordinate plane, and derived their algebraic form:

$$y = mx + c$$

This is the linear function $f(x) = mx + c$, first encountered in Week 1's discussion of functions. We now add one more layer of complexity: the *quadratic function*. Unlike the earlier approach (start geometric, derive algebraic), here we start algebraic and derive geometric properties.

2 Definition of a Quadratic Function

Quadratic Function

$$f(x) = ax^2 + bx + c, \quad a \neq 0$$

Why $a \neq 0$? If $a = 0$, the x^2 term vanishes and the expression reduces to $bx + c$ — a linear function, not quadratic.

Origin of the name: “Quadratic” derives from a term meaning *square*; a quadratic function is one involving the square of the variable, as seen in the ax^2 term.

The Three Terms

Term	Name	Role
ax^2	Quadratic term	Present only when $a \neq 0$; defines the curve
bx	Linear term	Survives when $a = 0$ (reduces to a line)
c	Constant term	The value of $f(x)$ when $x = 0$

3 Graphing a Quadratic Function: The Parabola

Parabola

The graph of any quadratic function is called a **parabola**.

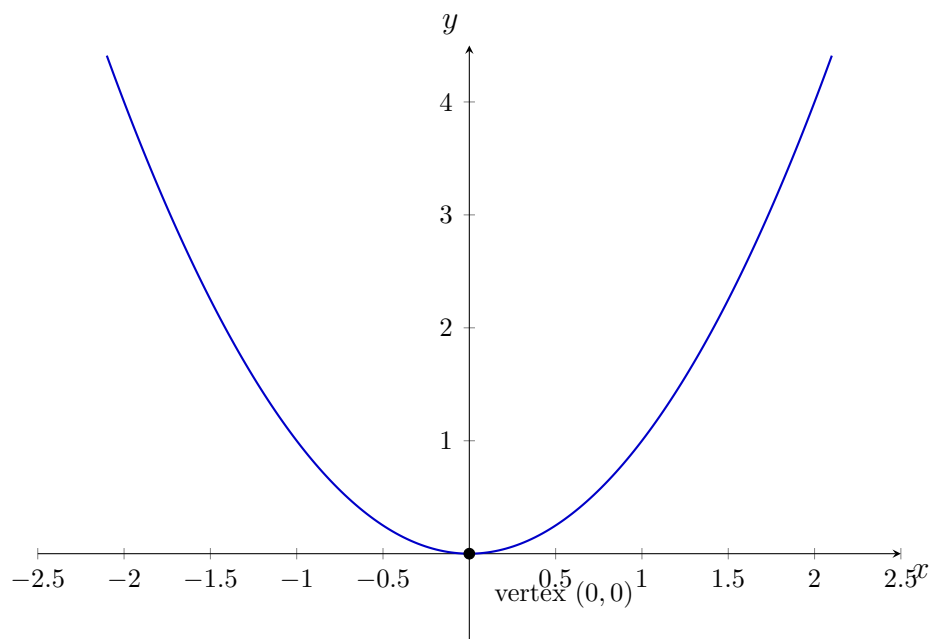
Method: exactly as with any function — choose values of x , compute $f(x)$, plot the ordered pairs $(x, f(x))$, and join them smoothly.

Example 1: $y = x^2$

Taking $a = 1$, $b = 0$, $c = 0$:

x	-2	-1	0	1	2
$y = x^2$	4	1	0	1	4

Plotting and joining these points smoothly gives the familiar upward-opening curve $y = x^2$.



Graph of $y = x^2$: an upward-opening parabola with vertex (and minimum) at the origin.

Example 2: $f(x) = x^2 + 2x + 1$

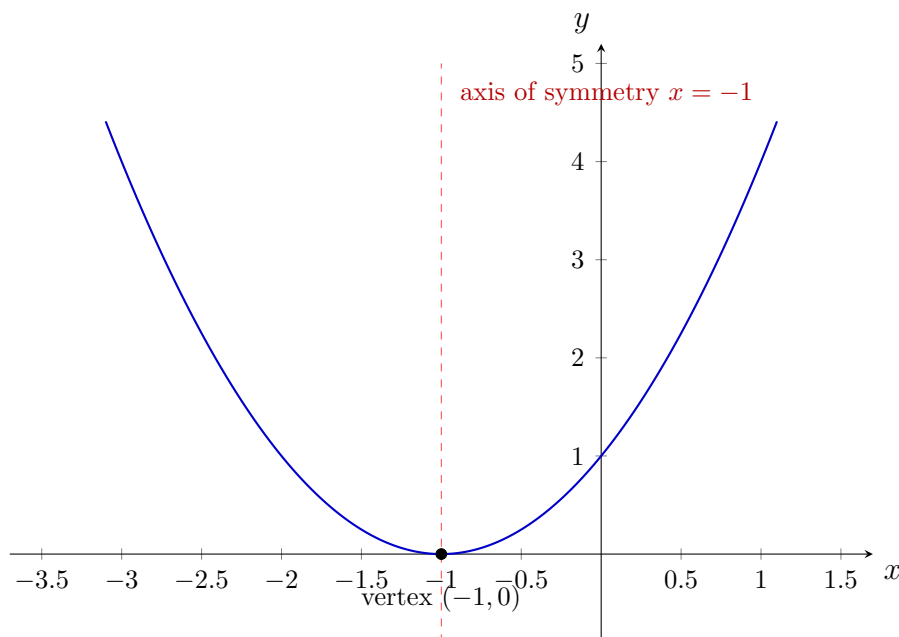
Three Steps to Graph Any Function

1. Generate a table of ordered pairs satisfying the function.
2. Plot those points on the coordinate plane.
3. Connect them with a smooth curve.

x	-2	-1	0	1
$f(x) = x^2 + 2x + 1$	1	0	1	4

Check for $x = -2$: $(-2)^2 + 2(-2) + 1 = 4 - 4 + 1 = 1$. ✓

An observation of symmetry: the point $x = 1$ gives $y = 4$; the point $x = -1$ gives $y = 0$. Moving two units further to $x = -3$ (mirroring the two units from $x = -1$ to $x = 1$) also gives $y = 4$. This hints at an underlying symmetry in the parabola, centered on the vertical line $x = -1$.



Graph of $f(x) = x^2 + 2x + 1$: symmetric about $x = -1$; vertex touches the x -axis at $(-1, 0)$.

4 Axis of Symmetry and Vertex

Axis of Symmetry

Every parabola $y = ax^2 + bx + c$ ($a \neq 0$) has a vertical line called the **axis of symmetry**. If the graph paper is folded along this line, the two halves of the parabola match exactly. Knowing the behaviour of the parabola on one side of the axis automatically gives the behaviour on the other side (a pure reflection).

Vertex

The point where the axis of symmetry meets the parabola is called the **vertex** of the parabola.

Three Important Quantities for $f(x) = ax^2 + bx + c$

- **Y-intercept:** set $x = 0 \Rightarrow y\text{-intercept} = c$.
- **Axis of symmetry:** $x = -\frac{b}{2a}$ (memorize for now; derivation requires algebraic tools covered later).
- **Vertex:** x -coordinate is $-\frac{b}{2a}$; the y -coordinate is found by substituting this x back into $f(x)$.

Note that c plays no role in the axis of symmetry — only a and b matter.

Worked Example: $f(x) = x^2 + 8x + 9$

Here $a = 1$, $b = 8$, $c = 9$.

Y-intercept: $c = 9 \Rightarrow (0, 9)$ lies on the parabola.

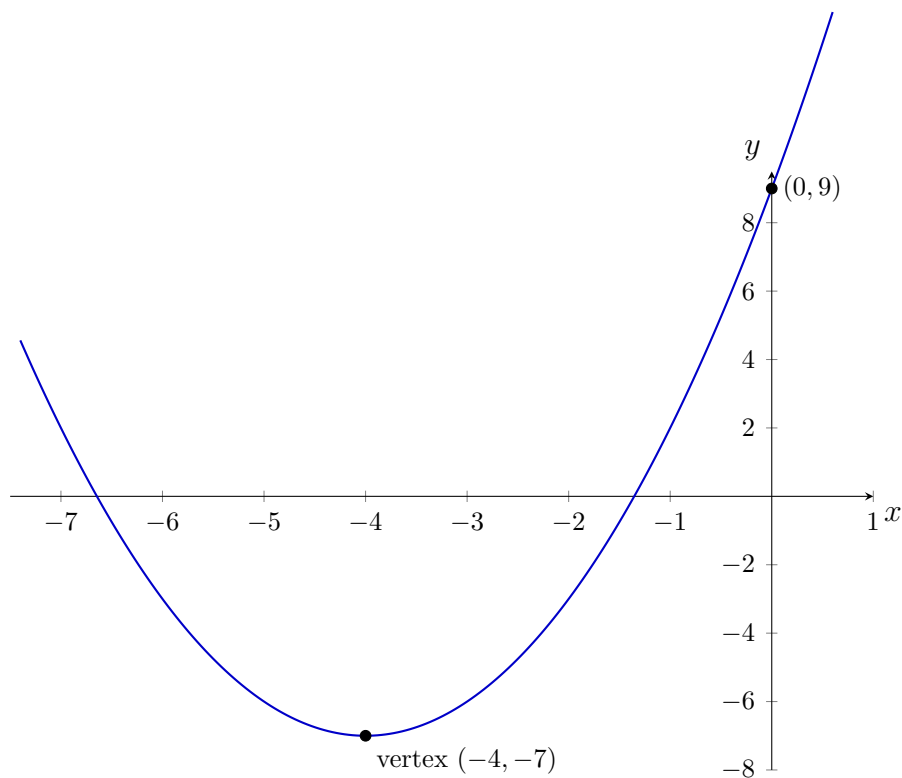
Axis of symmetry: $x = -\frac{b}{2a} = -\frac{8}{2} = -4$.

Vertex: $f(-4) = (-4)^2 + 8(-4) + 9 = 16 - 32 + 9 = -7 \Rightarrow (-4, -7)$.

Building the Table Around the Vertex

Since the parabola is symmetric about $x = -4$, points are chosen around the vertex:

x	-5	-4	-3	0
$f(x)$	-6	-7	-6	9



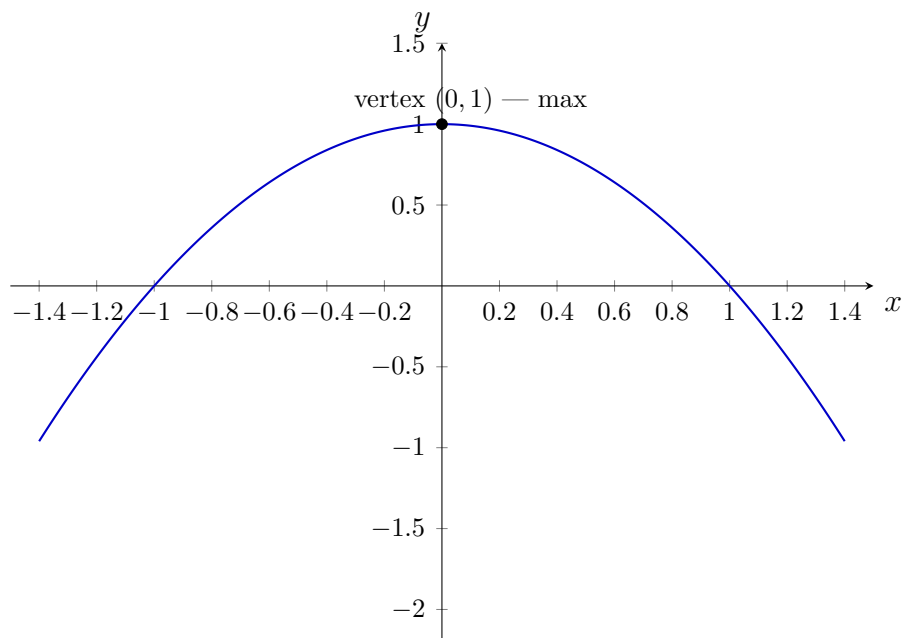
Graph of $f(x) = x^2 + 8x + 9$: upward-opening, vertex (minimum) at $(-4, -7)$, y -intercept at $(0, 9)$.

5 Opening Upward vs. Opening Downward

Example: $f(x) = -x^2 + 1$

Here $a = -1$, $b = 0$, $c = 1$. Y -intercept: 1. Axis of symmetry: $x = 0$ (the Y -axis, since $b = 0$). Vertex: $(0, 1)$. Since the axis of symmetry passes through the y -intercept, the vertex coincides with it.

x	-1	0	1
$f(x)$	0	1	0



Graph of $f(x) = -x^2 + 1$: since $a = -1 < 0$, the parabola opens downward with a maximum at $(0, 1)$.

The Role of the Sign of a

Whenever $a > 0$ (e.g. $a = 1$), the parabola opens **upward**. When $a < 0$ (e.g. $a = -1$), the parabola opens **downward**.

6 Maximum and Minimum Values

Extremum at the Vertex

The y -coordinate of the vertex is the minimum or maximum value attained by the quadratic function.

- If $a > 0$: the parabola opens upward and has a **minimum** value.
- If $a < 0$: the parabola opens downward and has a **maximum** value.

A quadratic function has either a minimum or a maximum — *never both*.

7 Range of a Quadratic Function

For $f(x) = ax^2 + bx + c$ ($a \neq 0$), the domain is always all of $\mathbb{R}$, but the range is restricted. Let $f_{\min}$ or $f_{\max}$ denote the vertex's y -value.

Range Formulas

$$a > 0 : \text{ Range} = \{ y \in \mathbb{R} : y \geq f_{\min} \}$$

$$a < 0 : \text{ Range} = \{ y \in \mathbb{R} : y \leq f_{\max} \}$$

In words: if the parabola opens upward, the range is all real numbers at or above the minimum; if it opens downward, all real numbers at or below the maximum.

8 Summary — Lecture 1

Quantity	Formula / Rule
General form	$f(x) = ax^2 + bx + c, a \neq 0$
Y-intercept	c
Axis of symmetry	$x = -b/(2a)$
Vertex	$(-\frac{b}{2a}, f(-\frac{b}{2a}))$
Opens upward, minimum	$a > 0$
Opens downward, maximum	$a < 0$
Range ($a > 0$)	$y \geq f_{\min}$
Range ($a < 0$)	$y \leq f_{\max}$

9 Key Takeaways — Lecture 1

1. A quadratic function has the form $f(x) = ax^2 + bx + c$ with $a \neq 0$; if $a = 0$ it degenerates into a linear function.
2. The graph is a parabola, drawn via the three-step method: table of values $\rightarrow$ plot points $\rightarrow$ join smoothly.
3. Every parabola has an axis of symmetry at $x = -b/(2a)$, and a vertex where this axis meets the curve; the vertex's y -coordinate is the min or max.
4. Knowing the y -intercept (c), axis of symmetry, and vertex makes plotting far easier.
5. The sign of a determines whether the parabola opens upward ($a > 0$, min) or downward ($a < 0$, max) — never both.
6. The domain is all of $\mathbb{R}$; the range is bounded on one side by the vertex's y -value.

Lecture 2 — Min/Max Problems & Introducing Slope of a Quadratic

A Worked Example, a Real-World Application, and a New Question

1 Worked Example: Domain, Range, Minimum/Maximum

Question

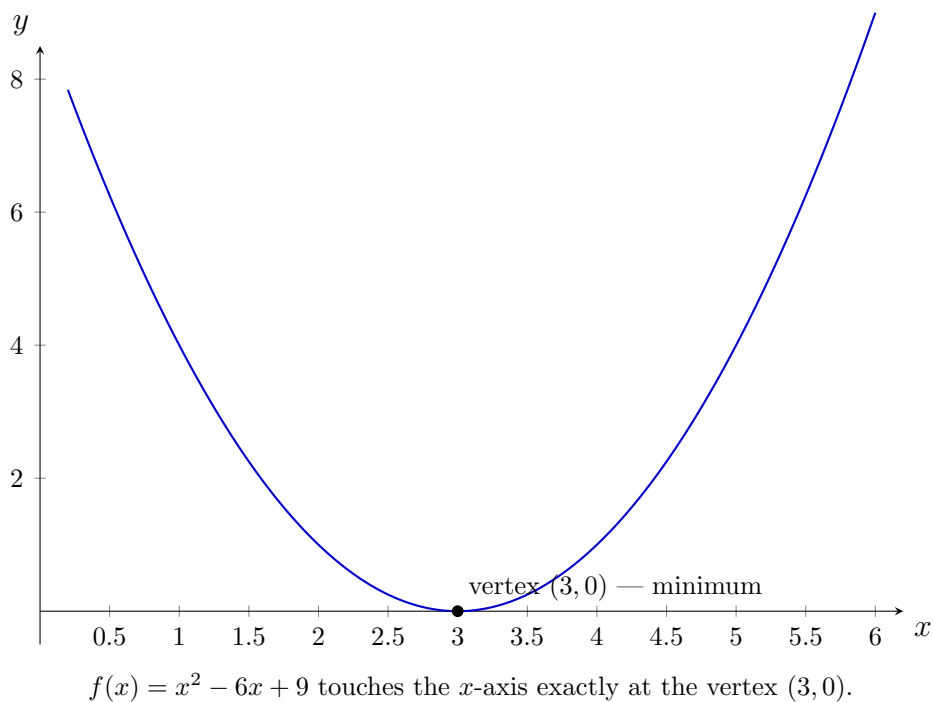
For $f(x) = x^2 - 6x + 9$, determine whether f has a minimum or maximum value (and find it), and state the domain and range.

Domain: all of $\mathbb{R}$. Coefficients: $a = 1$, $b = -6$, $c = 9$. Since $a > 0$, the parabola opens upward, so f has a **minimum**.

$$\text{Vertex } x: \quad x = -\frac{b}{2a} = -\frac{-6}{2(1)} = 3.$$

$$\text{Minimum value: } f(3) = 3^2 - 6(3) + 9 = 9 - 18 + 9 = 0.$$

$$\text{Range: } \{y \in \mathbb{R} : y \geq 0\}.$$



2 A Real-World Application: Maximizing Bus Revenue

Problem Setup

A tour bus in Chennai serves 500 customers/day at Rs 40 per person. For every Rs 4 hike in fare, the company loses 10 passengers/day. What fare maximizes income?

Let x = number of Rs 4 fare hikes. New fare: $40 + 4x$; remaining passengers: $500 - 10x$.

$$\text{Income}(x) = (500 - 10x)(40 + 4x) = -40x^2 + 1600x + 20000.$$

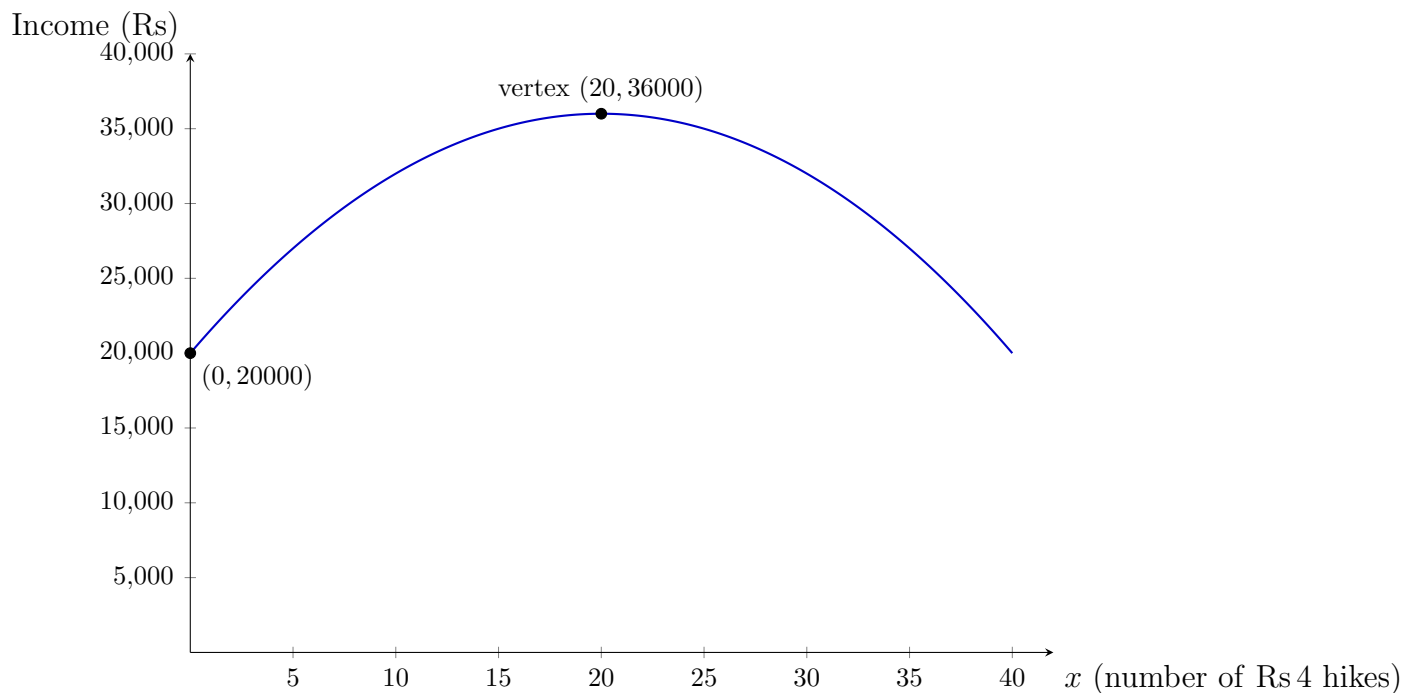
Here $a = -40$, $b = 1600$, $c = 20000$. Since $a < 0$, a maximum exists at the vertex:

$$x = -\frac{b}{2a} = -\frac{1600}{2(-40)} = \frac{1600}{80} = 20.$$

Substituting $x = 20$ gives maximum income Rs 36,000.

Comparing Before and After

Current income ($x = 0$): $500 \times 40 = \text{Rs } 20,000/\text{day}$. Recommended hike: $x = 20$ units = Rs 80 total. New fare: $40 + 80 = \text{Rs } 120/\text{person}$. New income: Rs 36,000/day, even though fewer passengers are served ($500 - 10(20) = 300$).



$$\text{Income}(x) = -40x^2 + 1600x + 20000: \text{ max income Rs } 36,000 \text{ at } x = 20.$$

This demonstrates how the vertex formula $x = -b/(2a)$ directly solves a practical business optimization problem.

3 A New Question: Slope of a Quadratic Function

Motivation. For a linear function $y = g(x) = mx + c$:

- The slope m is the ratio $\frac{\text{change in } y}{\text{change in } x}$.
- The slope is constant everywhere along the line.
- The slope equals $\tan(\theta)$, where θ is the angle of inclination with the positive x -axis.

Goal: For a quadratic function $f(x) = ax^2 + bx + c$ (with $a \neq 0$), how should the notion of “slope” be defined and computed?

Unlike a line, a parabola does not have a single constant slope — its steepness visibly changes from point to point. Near the vertex it is nearly flat, while far from the vertex it becomes steep. The natural starting point for this investigation is the prototype example $y = x^2$, to be explored in the next lecture.

4 Summary — Lecture 2

Task	Key Result
$f(x) = x^2 - 6x + 9$: min/max, range	Min = 0 at $x = 3$; range $y \geq 0$
Bus revenue $\text{Income}(x) = -40x^2 + 1600x + 20000$	Max = Rs 36,000 at $x = 20$
Optimal fare	Rs 120 per person
New question raised	Slope of a quadratic (not constant)

5 Key Takeaways — Lecture 2

1. To find the min/max, identify a, b, c ; the sign of a tells which extremum exists, and the vertex gives its value.
2. Real-world optimization can often be modeled as a quadratic, with the optimal decision at the vertex.
3. In the bus example, hiking the fare to Rs 120 maximizes income at Rs 36,000/day despite fewer passengers.
4. Unlike a line, a parabola’s steepness varies from point to point, motivating a generalized notion of slope.

Lecture 3 — Deriving the Slope of a Quadratic Function

From Secant Lines to $2ax + b$

1 Starting with the Prototype: $y = x^2$

Since $b = 0$, the Y -axis is the axis of symmetry, and the minimum value is 0 at $(0, 0)$.

x_i	−2	−1	0	1	2
$y_i = x_i^2$	4	1	0	1	4

2 First Attempt: Slope Between Consecutive Points

Since $\Delta x = 1$ between consecutive points, the slope $\frac{\Delta y}{\Delta x}$ reduces to $y_i - y_{i-1}$:

Interval	Difference $y_i - y_{i-1}$	Value
−1 to −2	$1 - 4$	−3
0 to −1	$0 - 1$	−1
1 to 0	$1 - 0$	1
2 to 1	$4 - 1$	3

A Key Caution

These differences are the slopes of straight lines connecting *pairs* of points on the curve — *not* the slope of the curve itself at any single point. A curve does not have one constant slope everywhere.

3 Second Attempt: The Tangent Line at a Point

Idea: at any point on the curve, draw the line that touches it at exactly that one point (a *tangent line*). Its slope is taken as the slope of the curve there.

Consider $(1.5, 2.25)$ on $y = x^2$. The tangent there has slope 3 — exactly the secant difference for the interval $x = 1$ to $x = 2$. Note 1.5 is the midpoint of 1 and 2.

- Midpoint of 0, 1 is 0.5 $\rightarrow$ tangent slope = 1.
- Midpoint of −1, 0 is −0.5 $\rightarrow$ tangent slope = −1.
- Midpoint of −2, −1 is −1.5 $\rightarrow$ tangent slope = −3.

In each case the tangent slope equals *twice* the midpoint's x -value.

Conclusion for $y = x^2$

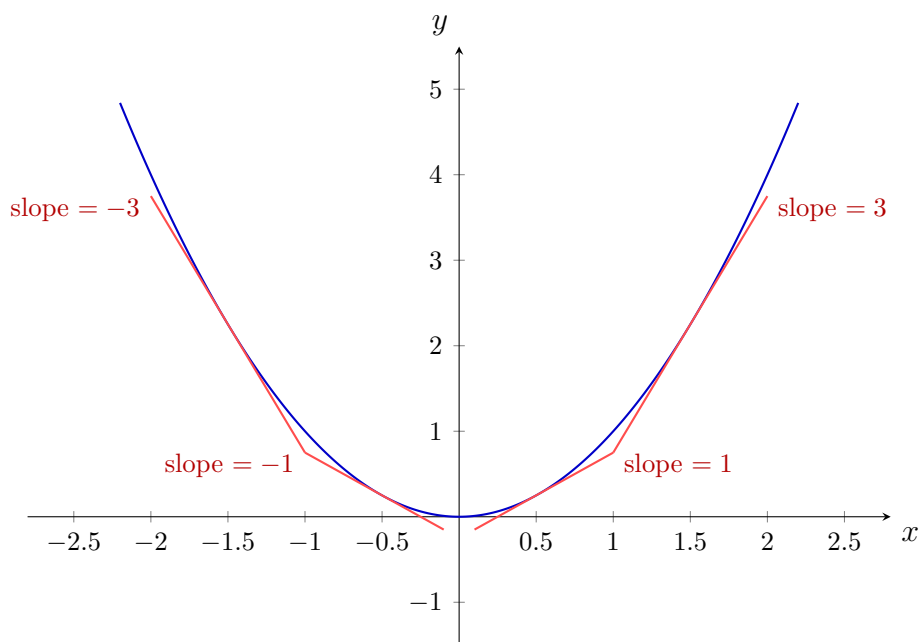
The slope of the curve at a point x is $\boxed{2x}$.

Verification at $x = 0$

Take symmetric points $x = 1$ and $x = -1$: the line joining $(-1, 1)$ and $(1, 1)$ is horizontal, slope 0 — consistent with $2x = 2(0) = 0$. ✓

Special Recipe (for Quadratic function only)

To find the slope at a point z : pick two symmetric points $z + u$ and $z - u$ around it, join them with a straight line, and compute that line's slope.



Tangent lines to $y = x^2$ at $x = \pm 0.5, \pm 1.5$: the tangent slope at x equals $2x$ (values 3, 1, -1, -3 shown).

4 Generalizing to $f(x) = ax^2 + bx + c$

Take $x = -2, -1, 0, 1, 2$ and evaluate:

$$f(-2) = 4a - 2b + c, \quad f(-1) = a - b + c, \quad f(0) = c, \quad f(1) = a + b + c, \quad f(2) = 4a + 2b + c.$$

Consecutive differences equal $2a$ times the midpoint plus b :

Midpoint	Difference	Matches
-1.5	$-3a + b$	$2a(-1.5) + b$
-0.5	$-a + b$	$2a(-0.5) + b$
0.5	$a + b$	$2a(0.5) + b$
1.5	$3a + b$	$2a(1.5) + b$

Slope of a Quadratic Function

Slope of $f(x) = ax^2 + bx + c$ at a point x is $2ax + b$.

This depends only on a and b — not on c , since c is just a vertical shift that does not affect steepness.

5 Why This Explains the Vertex Formula

The slope $2ax + b$ measures the rate of change of y w.r.t. x . At the min or max, this must be zero (the curve is momentarily flat):

$$2ax + b = 0 \Rightarrow x = -\frac{b}{2a}.$$

This is exactly the vertex formula — now *justified* rather than memorized.

6 Slope of a Line vs. Slope of a Quadratic

	Line $y = mx + c$	Quadratic $f(x) = ax^2 + bx + c$
Slope	m (constant)	$2ax + b$ (varies with x)
Depends on	only m	a and x (and b); not c
Zero slope	never (unless $m = 0$)	at $x = -b/2a$ (the vertex)

7 Summary — Lecture 3

Concept	Result
Slope of $y = x^2$ at x	$2x$
Slope of $f(x) = ax^2 + bx + c$ at x	$2ax + b$
Slope = 0 at	$x = -b/2a$ (vertex)
Method used	Tangent line via symmetric midpoint secants

8 Key Takeaways — Lecture 3

1. Unlike a line, a curve's slope is not constant — it is defined at each point using the tangent line.

2. For $y = x^2$, the slope at x equals $2x$.
3. Generalizing to $f(x) = ax^2 + bx + c$ gives slope $2ax + b$.
4. The slope depends on a and b but not c .
5. Setting $2ax + b = 0$ recovers $x = -b/2a$, explaining the location of the min/max.

Lecture 4 — Quadratic Equations & Finding Roots by Graphing

Standard Form, Roots as X-Intercepts, and the Number of Real Roots

1 From Quadratic Function to Quadratic Equation

Quadratic Equation

If a quadratic function is set equal to a value, the result is a **Quadratic Equation**. E.g. $ax^2 + bx + c = 0$ and $ax^2 + bx + c = 5$ (provided $a \neq 0$).

Standard Form

If the coefficients are integers and the equation is set equal to 0:

$$ax^2 + bx + c = 0, \quad a, b, c \in \mathbb{Z}, \quad a \neq 0.$$

2 Roots of a Quadratic Equation

Root

The solutions of $ax^2 + bx + c = 0$ — the values of x for which the expression equals 0 — are the **roots** of the equation.

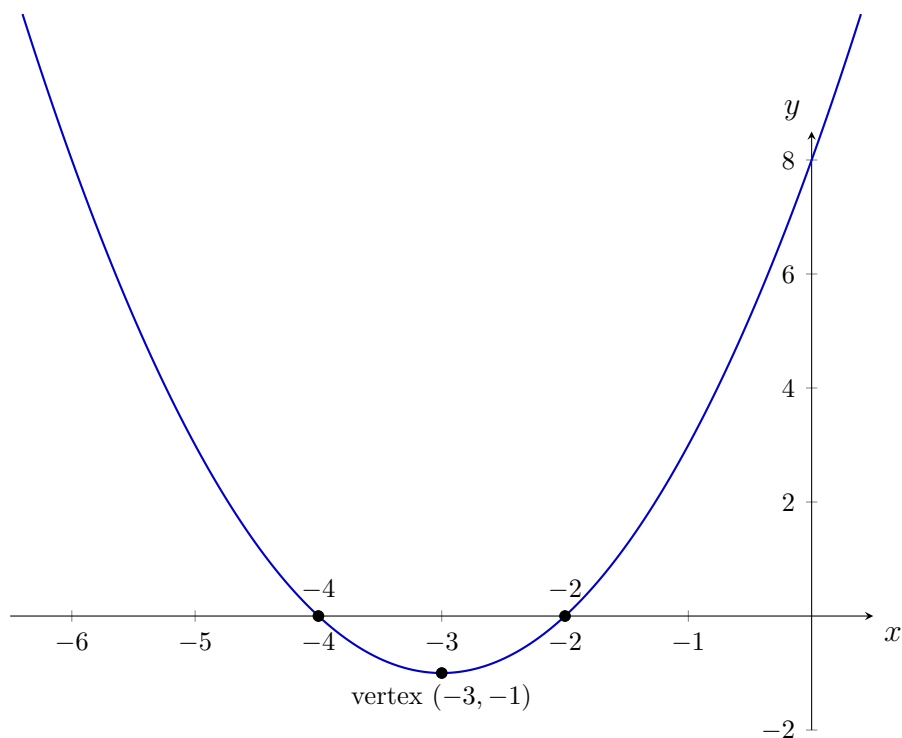
The zeroes of $f(x) = ax^2 + bx + c$ are precisely its x -intercepts, which are exactly the roots of the associated equation. Method: plot the function and read off where the curve crosses the x -axis.

3 Worked Example 1: $x^2 + 6x + 8 = 0$

$a = 1$, $b = 6$, $c = 8$. Y -intercept: 8. Axis: $x = -\frac{6}{2} = -3$. Since $a > 0$ and the vertex dips below the x -axis, the curve crosses at two points. Try $x = -2$: $(-2)^2 + 6(-2) + 8 = 4 - 12 + 8 = 0$. By symmetry about $x = -3$, the other root is $x = -4$. Check: $(-4)^2 + 6(-4) + 8 = 16 - 24 + 8 = 0$. ✓

Conclusion

Roots: $x = -4$ and $x = -2$ — two distinct real roots. Vertex $(-3, -1)$.



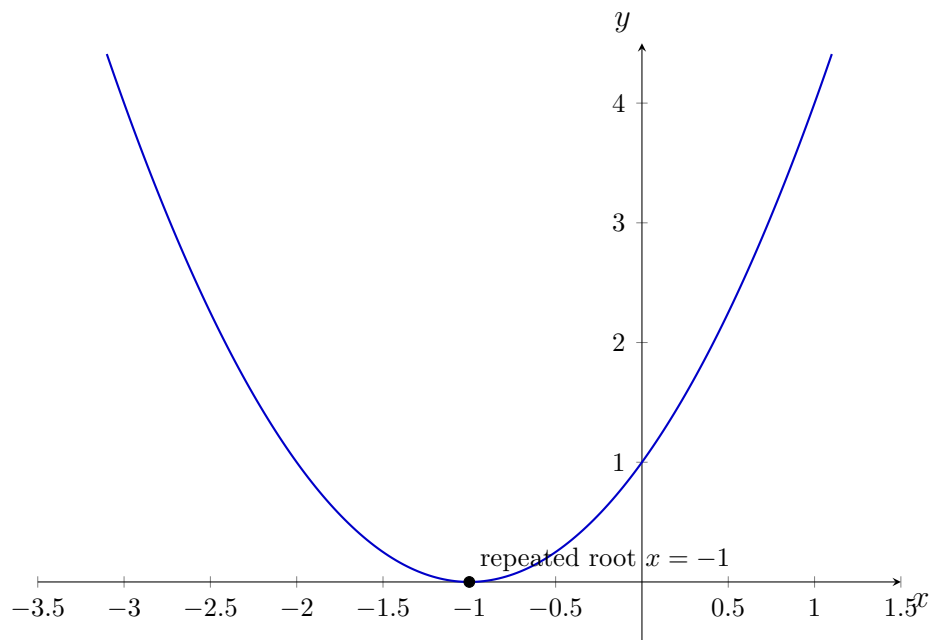
$x^2 + 6x + 8 = 0$: vertex below the x -axis $\Rightarrow$ two real roots.

4 Worked Example 2: $x^2 + 2x + 1 = 0$

$a = 1$, $b = 2$, $c = 1$. Axis: $x = -\frac{2}{2} = -1$. Vertex value: $f(-1) = 1 - 2 + 1 = 0$.

A Special Case: Repeated Root

The vertex lies on the x -axis. Since $a > 0$, the curve touches the axis exactly at $x = -1$ and cannot cross elsewhere. There is one real root, $x = -1$, occurring twice (a **repeated** / **double root**).



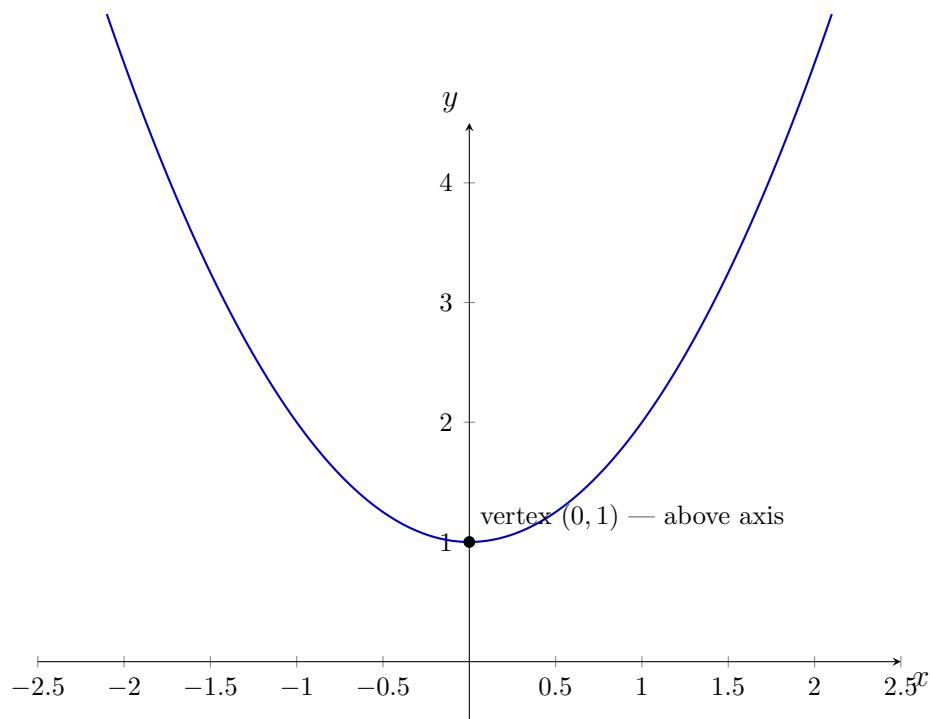
$x^2 + 2x + 1 = 0$: vertex touches the x -axis $\Rightarrow$ one repeated root.

5 Worked Example 3: $x^2 + 1 = 0$

$a = 1$, $b = 0$, $c = 1$. Axis: $x = 0$. Vertex: $(0, 1)$, the minimum. Since the minimum value is $1 > 0$, the curve never dips to the x -axis.

Conclusion

This equation has **no real roots**.



$x^2 + 1 = 0$: vertex above the x -axis and $a > 0 \Rightarrow$ no real roots.

6 General Classification of Roots via the Vertex

$a > 0$ (opens upward, minimum at vertex)

- Vertex y -coordinate negative $\Rightarrow$ two real roots.
- Vertex y -coordinate zero $\Rightarrow$ one repeated real root.
- Vertex y -coordinate positive $\Rightarrow$ no real roots.

$a < 0$ (opens downward, maximum at vertex)

- Vertex y -coordinate positive $\Rightarrow$ two real roots.
- Vertex y -coordinate zero $\Rightarrow$ one repeated real root.
- Vertex y -coordinate negative $\Rightarrow$ no real roots.

7 Summary — Lecture 4

Example	Vertex	Sign of y -coordinate of vertex	Real roots
$x^2 + 6x + 8 = 0$	$(-3, -1)$	Negative	Two: $x = -4, -2$
$x^2 + 2x + 1 = 0$	$(-1, 0)$	Zero	One (repeated): $x = -1$
$x^2 + 1 = 0$	$(0, 1)$	Positive	None

8 Key Takeaways — Lecture 4

1. A quadratic equation arises from setting a quadratic function equal to a value; standard form is $ax^2 + bx + c = 0$ with $a, b, c \in \mathbb{Z}$, $a \neq 0$.
2. The roots equal the x -intercepts (zeroes) of the associated function.
3. Once the axis and one root are known, symmetry gives the second root.
4. The number of real roots is determined by the sign of the vertex's y -coordinate relative to the sign of a .
5. This graphing technique gives a purely visual method for the number and location of roots.

Lecture 5 — Revisiting Slope of a Line and Generalizing to Quadratics

From Symmetric Difference Quotients to $2ax + b$

1 Recap: Slope of a Line

Slope of a Line

$$\text{slope} = \frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\Delta y}{\Delta x}.$$

Slope measures the sensitivity of the output to a change in input — the *rate of change*. For a line this is constant everywhere, giving $y = mx + b$.

2 Why Generalize the Notion of Slope?

For a line the rate of change is trivially constant. But what is the rate of change of a function whose graph is not a straight line — e.g. a parabola? The notion of slope must be generalized to apply to any function.

3 A General Definition: The Symmetric Difference Quotient

Take a small interval of size h on either side of x_0 , giving symmetric points $x_0 - h$ and $x_0 + h$; apply rise-over-run to this pair.

Slope of a Function at a Point (symmetric limit is defined for QE here, not for any function in general)

$$\text{slope at } x_0 = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0 - h)}{(x_0 + h) - (x_0 - h)} = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0 - h)}{2h}.$$

4 Verifying the Definition on a Line

For $f(x) = mx + b$:

$$\text{slope at } x_0 = \lim_{h \rightarrow 0} \frac{[m(x_0 + h) + b] - [m(x_0 - h) + b]}{2h} = \lim_{h \rightarrow 0} \frac{2mh}{2h} = m.$$

Check

The formula recovers slope = m , matching the known constant slope of a line. The general definition is consistent with the familiar case.

5 Worked Example: Slope of $y = x^2$

$$\text{slope at } x_0 = \lim_{h \rightarrow 0} \frac{(x_0 + h)^2 - (x_0 - h)^2}{2h}.$$

Expanding: $(x_0 + h)^2 - (x_0 - h)^2 = 4hx_0$, so

$$\text{slope at } x_0 = \lim_{h \rightarrow 0} \frac{4hx_0}{2h} = \lim_{h \rightarrow 0} 2x_0 = 2x_0.$$

Conclusion

The slope of $y = x^2$ at a point x is $2x$. This defines a new function $g(x) = 2x$.

6 Generalizing to $f(x) = ax^2 + bx + c$

$$\begin{aligned} \text{slope at } x_0 &= \lim_{h \rightarrow 0} \frac{[a(x_0 + h)^2 + b(x_0 + h) + c] - [a(x_0 - h)^2 + b(x_0 - h) + c]}{2h} \\ &= \lim_{h \rightarrow 0} \frac{a(4hx_0) + b(2h)}{2h} = \lim_{h \rightarrow 0} \frac{2h(2ax_0 + b)}{2h} = 2ax_0 + b. \end{aligned}$$

Slope of a Quadratic Function

For $f(x) = ax^2 + bx + c$, $a \neq 0$, the slope at a point x is $g(x) = 2ax + b$.

Consistency Check

Setting $a = 1, b = 0$ gives $g(x) = 2x$; and if $a = 0$ (so f is the line $bx + c$), the formula reduces to $g(x) = b$ — the constant slope of a line.

7 Connecting Slope to the Vertex

Setting $g(x) = 0$:

$$2ax + b = 0 \Rightarrow x = -\frac{b}{2a} \quad (a \neq 0).$$

Why This Matters

$x = -\frac{b}{2a}$ is exactly the vertex — where the quadratic attains its min or max. The rate of change is positive on one side and negative on the other, passing through zero exactly at the vertex. An early preview of calculus.

8 Summary — Lecture 5

Function	Slope at a point x
Line: $f(x) = mx + b$	m (constant)
$y = x^2$	$2x$
General quadratic: $f(x) = ax^2 + bx + c$	$2ax + b$

Setting $2ax + b = 0$ gives $x = -\frac{b}{2a}$ — the vertex.

9 Key Takeaways — Lecture 5

1. The slope of a line is $\frac{\Delta y}{\Delta x}$, constant everywhere.
2. For $y = x^2$ it gives $2x$; for $f(x) = ax^2 + bx + c$ it gives $2ax + b$.
3. Setting the slope to zero gives $x = -b/2a$ — the vertex.

Lecture 6 — Course Summary: Graphing, Slope, and the Vertex

Putting Graphing, Slope, and Extrema Together

1 Recap: From Lines to Quadratic Functions

The Analogy

A non-vertical line has the form $f(x) = mx + b$ with $m \in \mathbb{R}$. In the same way, a quadratic function is $f(x) = ax^2 + bx + c$, $a \neq 0$, where $a \neq 0$ ensures the x^2 term keeps its relevance.

2 The Three-Point Recipe for Graphing

Cookbook Recipe

To graph $f(x) = ax^2 + bx + c$:

1. Find the axis of symmetry, $x = -\frac{b}{2a}$.
2. Find the y -intercept by evaluating $f(0)$.
3. Pick an arbitrary point x_1 , compute $y_1 = f(x_1)$, then use symmetry to get a second point x_2 with $y_2 = y_1$.

3 Worked Example: Graphing $f(x) = x^2 + 4x + 4$

Here $a = 1$, $b = 4$, $c = 4$.

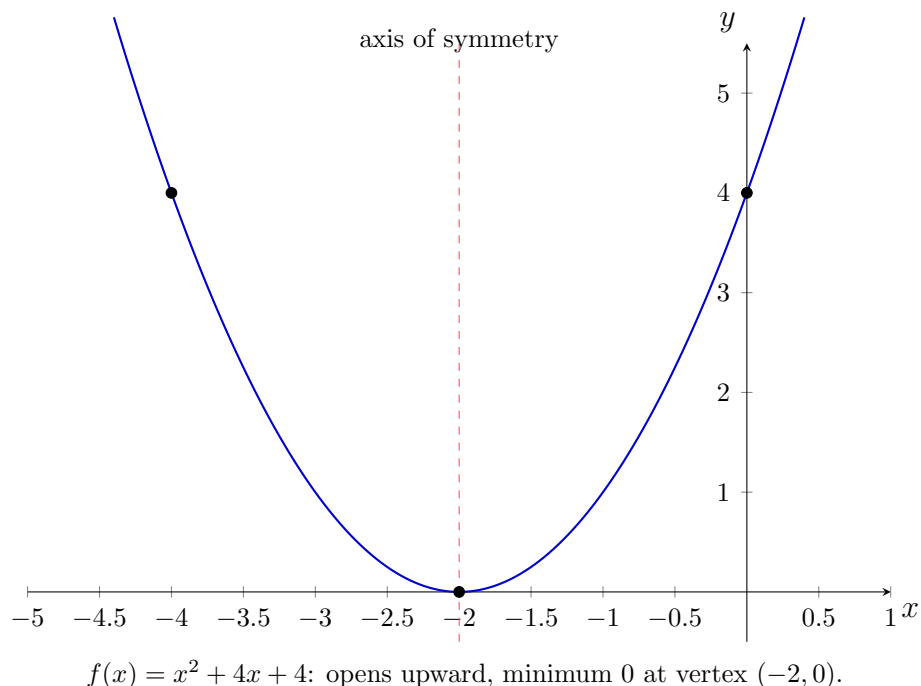
$$\text{Axis: } x = -\frac{4}{2(1)} = -2.$$

$$Y\text{-intercept: } f(0) = 4 \Rightarrow (0, 4).$$

$$\text{Symmetric point: } f(-4) = 16 - 16 + 4 = 4 \Rightarrow (-4, 4).$$

Points Found

$(-4, 4)$, $(-2, 0)$ (on the axis of symmetry), and $(0, 4)$ — three points, symmetric about $x = -2$, are enough to sketch the full curve.



4 Recap: Slope of a Quadratic Function

Slope via Symmetric Difference

$$g(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x-h)}{2h}, \quad \text{giving} \quad g(x) = 2ax + b.$$

For the example, $g(x) = 2x + 4$; setting $g(x) = 0$ gives $x = -2$, matching the axis.

Consistency Check

In general $x = -\frac{b}{2a}$ is always the vertex — the point where the slope is zero, and where the function changes its monotonicity.

5 Minimum vs. Maximum: The Role of the Sign of a

For $f(x) = x^2 + 4x + 4$, the vertex is $(-2, 0)$, so the minimum value is 0. Multiplying f by -1 reflects the graph across the x -axis: a minimum of 0 becomes a maximum of 0.

General Rule

For $f(x) = ax^2 + bx + c$: if $a > 0$ the function attains a **minimum** (at the vertex); if $a < 0$ it attains a **maximum**.

6 Summary — Lecture 6

Task	Key Result
Graphing recipe	Axis of symmetry, y -intercept, one symmetric pair
Axis of symmetry	$x = -b/2a$
Example $f(x) = x^2 + 4x + 4$	Vertex $(-2, 0)$
Slope of a quadratic	$g(x) = 2ax + b$
Vertex from slope	$g(x) = 0 \Rightarrow x = -b/2a$
Min. vs. max.	$a > 0 \Rightarrow \text{min}; a < 0 \Rightarrow \text{max}$

7 Key Takeaways — Lecture 6

1. A quadratic is graphed via a three-point recipe: axis $x = -b/2a$, y -intercept $f(0)$, and one symmetric pair.
2. For $f(x) = x^2 + 4x + 4$: axis $x = -2$, y -intercept $(0, 4)$, mirror point $(-4, 4)$.
3. The slope is $g(x) = 2ax + b$; setting it to zero recovers $x = -b/2a$.
4. A quadratic attains either a minimum or a maximum at its vertex, never both.
5. The sign of a decides which: $a > 0$ min, $a < 0$ max.

Lecture 7 — Solving Quadratic Equations by Factoring: The Intercept Form

Intercept Form, the FOIL Method, and Converting Between Forms

1 Recap: Roots by Graphing

Previously, roots were found by graphing and reading off x -axis crossings. This lecture introduces an algebraic alternative: the **factoring method**, built around the **intercept form**.

2 The Intercept Form

Intercept Form

A quadratic function is in intercept form when written as

$$y = a(x - p)(x - q),$$

where p, q are precisely the x -intercepts (roots) of the function.

3 Worked Example: Reading Off a Third Point

Given $y = 3(x - 1)(x - 5)$ with roots 1 and 5, the axis of symmetry is the average:

Axis of Symmetry from Two Roots

$$x = \frac{p + q}{2} = \frac{1 + 5}{2} = 3.$$

Substituting $x = 3$: $y = 3(3 - 1)(3 - 5) = 3(2)(-2) = -12$.

Three Points Found

$(1, 0)$, $(5, 0)$, and $(3, -12)$ — enough to sketch the full parabola.

4 Converting Intercept Form to Standard Form: FOIL

FOIL Method

To multiply $(ax + b)(cx + d)$, sum the products of: **F**irst $(ax \cdot cx)$, **O**uter $(ax \cdot d)$, **I**nnner $(b \cdot cx)$, **L**ast $(b \cdot d)$:

$$(ax + b)(cx + d) = acx^2 + (ad + bc)x + bd.$$

Key Observation

Comparing $acx^2 + (ad + bc)x + bd$: $(ac)(bd) = abcd$ and $(ad)(bc) = abcd$. The product of the x^2 -coefficient and constant equals the product of the two “split” terms that add up to the x -coefficient — the key to converting between forms in both directions.

5 Worked Example: Standard Form from Given Roots

Roots $\frac{2}{3}$ and -4 . Intercept form: $(x - \frac{2}{3})(x + 4) = 0$. Applying FOIL:

$$x^2 + (4 - \frac{2}{3})x + (-\frac{2}{3} \cdot 4) = x^2 + \frac{10}{3}x - \frac{8}{3} = 0.$$

Multiplying through by 3:

Result

$$3x^2 + 10x - 8 = 0.$$

6 Worked Example: Intercept Form from Standard Form

Convert $5x^2 - 13x + 6$. Match $(ax + b)(cx + d)$: $ac = 5$, $ad + bc = -13$, $bd = 6$.

Applying the Key Observation

$(ac)(bd) = 30$, so the split-terms ad, bc multiply to 30 and add to -13 . Since $(-10)(-3) = 30$ and $(-10) + (-3) = -13$ ✓, take $ad = -10$, $bc = -3$.

Rewriting the middle term:

$$5x^2 - 10x - 3x + 6 = 5x(x - 2) - 3(x - 2) = (5x - 3)(x - 2).$$

Dividing $5x - 3$ by 5 and pulling the 5 out front:

Result

$$5x^2 - 13x + 6 = 5 \left(x - \frac{3}{5}\right)(x - 2).$$

7 Further Examples

Example: $x^2 = 8x$. Rearranging: $x^2 - 8x = 0$. Factor out x : $x(x - 8) = 0 \Rightarrow x = 0, 8$.

Example: $x^2 - 4x + 4 = 0$ (**repeated root**). Here $abcd = 4$, $ad + bc = -4$; both split-terms are -2 :

$$x^2 - 2x - 2x + 4 = x(x - 2) - 2(x - 2) = (x - 2)^2 = 0 \Rightarrow x = 2.$$

Example: $x^2 - 25 = 0$ (**difference of squares**). Here $abcd = -25$, $ad + bc = 0$; use $+5$ and -5 :

$$x^2 - 5x + 5x - 25 = x(x - 5) + 5(x - 5) = (x + 5)(x - 5) = 0 \Rightarrow x = \pm 5.$$

Takeaway

Once rewritten in intercept form, the roots are immediately visible — converting to intercept form and solving the equation are, in effect, the same task.

8 Summary — Lecture 7

Concept	Key Result
Intercept form	$y = a(x - p)(x - q)$, with p, q the x -intercepts
Axis of symmetry from roots	$x = (p + q)/2$
FOIL expansion	$(ax + b)(cx + d) = acx^2 + (ad + bc)x + bd$
Key observation	$(ac)(bd) = (ad)(bc) = abcd$
Roots $\frac{2}{3}, -4$	Standard form: $3x^2 + 10x - 8 = 0$
$5x^2 - 13x + 6$	Intercept form: $5(x - \frac{3}{5})(x - 2)$
$x^2 - 4x + 4 = 0$	Repeated root $x = 2$
$x^2 - 25 = 0$	Roots $x = \pm 5$

9 Key Takeaways — Lecture 7

1. The intercept form $y = a(x - p)(x - q)$ directly displays the roots p, q .
2. Given two roots, the axis of symmetry is their average $x = (p + q)/2$.

Lecture 8 — Solving Quadratic Equations by Completing the Square

A New Method, and a Preview of the Quadratic Formula

1 Recap: The Factoring Method

For $x^2 + 10x - 24$: $abcd = -24$, $ad + bc = 10$; the split $12, -2$ gives

$$x^2 + 12x - 2x - 24 = x(x + 12) - 2(x + 12) = (x + 12)(x - 2),$$

so roots $x = -12$ and $x = 2$.

2 A New Idea: Completing the Square

Motivating Question

The factoring method depends on the constant -24 being easy to factor. What if we simply replace and work around it directly?

Rewrite $x^2 + 10x - 24$ as $x^2 + 10x = 24$ and ask what can be added to make the LHS a perfect square.

Perfect Square Form

$$(x + a)^2 = x^2 + 2ax + a^2.$$

Comparing $x^2 + 10x$ with $x^2 + 2ax$: $2a = 10 \Rightarrow a = 5 \Rightarrow a^2 = 25$.

3 Worked Example: $x^2 + 10x - 24 = 0$

Rewrite as $x^2 + 10x = 24$, then add $a^2 = 25$ to both sides:

$$x^2 + 10x + 25 = 24 + 25 = 49.$$

Recognizing the Perfect Squares

The LHS is $(x + 5)^2$; the RHS $49 = 7^2$:

$$(x + 5)^2 = 7^2.$$

Two Cases, Not Four

Both $+(x+5) = \pm 7$ and $-(x+5) = \pm 7$ reduce to the same pair. So it suffices to consider $x + 5 = \pm 7$, giving $x = -5 + 7 = 2$ and $x = -5 - 7 = -12$.

Roots

$x = 2$ and $x = -12$ — exactly matching the roots found by factoring.

4 When Is This Method Useful?

Advantage

Completing the square never relies on the constant term being easy to factor. It works by the same add-and-subtract procedure regardless of how “awkward” the numbers are.

A Limitation

If, after adding a^2 , the RHS turns out negative, this method cannot produce a real solution — the square root of a negative number is undefined in the reals. Addressed in upcoming lectures, along with the quadratic formula.

5 Summary — Lecture 8

Step	Action
1	Move the constant to the RHS: $x^2 + 10x = 24$
2	Match $2a$ with the x -coefficient: $2a = 10 \Rightarrow a = 5$, $a^2 = 25$
3	Add a^2 to both sides: $x^2 + 10x + 25 = 49$
4	Recognize both sides as perfect squares: $(x + 5)^2 = 7^2$
5	Solve $x + 5 = \pm 7$: roots $x = 2, -12$

6 Key Takeaways — Lecture 8

1. Completing the square rewrites $x^2 + bx = k$ by adding $a^2 = (b/2)^2$ to both sides, turning the LHS into $(x + a)^2$.
2. Once both sides are perfect squares, the equation reduces to two cases $x + a = \pm\sqrt{k'}$ rather than four.
3. Applied to $x^2 + 10x - 24 = 0$, it gives $x = 2, -12$ — identical to factoring.
4. Unlike factoring, it does not depend on an easy integer split of the constant.

5. If the RHS becomes negative, there is no real solution - – in that case we get complex roots.

Lecture 9 — The Quadratic Formula, Discriminant, and Axis of Symmetry

Completing the Square Generalized, and Why $x = -b/2a$ Works

1 Completing the Square When the RHS Isn't a Perfect Square

Consider $x^2 - 4x + 4 = 32$, i.e. $(x - 2)^2 = 32$, but 32 is not a perfect square.

The Same Method Still Works

Take $\pm$ the square root: $x - 2 = \pm\sqrt{32}$. Since $32 = 16 \times 2$, $\sqrt{32} = 4\sqrt{2}$.

Roots (Irrational)

$$x = 2 \pm 4\sqrt{2}.$$

These are irrational roots, since $\sqrt{2}$ is irrational.

2 Deriving the Quadratic Formula

Start with $ax^2 + bx + c = 0$ ($a \neq 0$). Divide by a and move the constant:

$$x^2 + \frac{b}{a}x = -\frac{c}{a}.$$

Matching $\frac{b}{a}x$ with $2\alpha x$ gives $\alpha = \frac{b}{2a}$, so $\alpha^2 = \frac{b^2}{4a^2}$. Add to both sides:

$$x^2 + \frac{b}{a}x + \frac{b^2}{4a^2} = \frac{b^2}{4a^2} - \frac{c}{a}.$$

Recognizing the Perfect Square

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}, \quad x + \frac{b}{2a} = \pm \frac{\sqrt{b^2 - 4ac}}{2a}.$$

The Quadratic Formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

The quantity $b^2 - 4ac$ under the square root is called the **discriminant**.

3 The Discriminant: What It Discriminates

Why “Discriminant”?

The discriminant $b^2 - 4ac$ determines the type of roots:

- $b^2 - 4ac > 0$: two real roots.
- $b^2 - 4ac = 0$: exactly one repeated real root.
- $b^2 - 4ac < 0$: no real roots (the roots are complex).

When a, b, c are rational, the discriminant also distinguishes rational from irrational roots:

Discriminant $b^2 - 4ac$	Further Condition	Roots
> 0	a perfect square	Two real, rational roots
> 0	not a perfect square	Two real, irrational roots
$= 0$	—	One real, repeated root
< 0	—	No real roots

A Caveat on Rationality

The rational/irrational classification assumes $a, b, c \in \mathbb{Q}$. If they are merely real, the count of real roots still holds, but nothing can be said about rationality. For instance, $(x - \pi)^2 = 0$ has discriminant 0 but repeated root π , which is irrational.

4 Worked Examples: Classifying Roots via the Discriminant

Example 1: $9x^2 - 12x + 4 = 0$. $a = 9, b = -12, c = 4$:

$$b^2 - 4ac = (-12)^2 - 4(9)(4) = 144 - 144 = 0.$$

Conclusion

Discriminant $= 0 \Rightarrow$ one real, repeated root.

Example 2: $2x^2 + 16x + 33 = 0$. $a = 2, b = 16, c = 33$:

$$b^2 - 4ac = 16^2 - 4(2)(33) = 256 - 264 = -8.$$

Conclusion

Discriminant $= -8 < 0 \Rightarrow$ no real roots.

5 Comparing the Four Solution Methods

Method	Best Used When	Limitation
Graphing	Verifying a result; occasionally finding integer roots	Imprecise unless roots are integers
Factoring	The constant factors into nice integers	Difficult if factors aren't easily visible
Completing the Square	Almost always; convenient when b is even	Leads into complex numbers if RHS negative
Quadratic Formula	Always — derived from completing the square	None for this course; works for any real a, b, c

Takeaway

Completing the square and the quadratic formula both work in every case, regardless of whether the coefficients are rational, irrational, or otherwise “unfriendly.”

6 Why $x = -b/2a$ Is the Axis of Symmetry

Factor a out of the first two terms and complete the square:

$$f(x) = a \left(x^2 + \frac{b}{a}x \right) + c = a \left(x + \frac{b}{2a} \right)^2 - \frac{b^2}{4a} + c.$$

Vertex Form

$$f(x) = a \left(x + \frac{b}{2a} \right)^2 + \left(c - \frac{b^2}{4a} \right).$$

The bracketed constant does not depend on x .

The expression is symmetric about the line where the squared quantity vanishes:

$$x + \frac{b}{2a} = 0 \Rightarrow x = -\frac{b}{2a}.$$

The additive constant only shifts the graph vertically.

Conclusion

The axis of symmetry of $f(x) = ax^2 + bx + c$ is $x = -\frac{b}{2a}$ — not an assumption, but a direct consequence of vertex form.

7 Summary — Lecture 9

Concept	Result
Completing square, non-perfect-square RHS	Roots may be irrational, e.g. $2 \pm 4\sqrt{2}$
Quadratic formula	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
Discriminant	$b^2 - 4ac$
$b^2 - 4ac > 0$, perfect square	Two real, rational roots
$b^2 - 4ac > 0$, not perfect square	Two real, irrational roots
$b^2 - 4ac = 0$	One repeated real root
$b^2 - 4ac < 0$	No real roots
Vertex form	$a \left(x + \frac{b}{2a}\right)^2 + \left(c - \frac{b^2}{4a}\right)$
Axis of symmetry (derived)	$x = -b/2a$

8 Key Takeaways — Lecture 9

1. When the RHS is not a perfect square, completing the square still works — it produces irrational roots such as $x = 2 \pm 4\sqrt{2}$.
2. Applying it to $ax^2 + bx + c = 0$ derives the quadratic formula.
3. The discriminant $b^2 - 4ac$ determines the number and type of roots.
4. Each method has its place: graphing for verification, factoring when numbers are friendly, and completing-the-square / formula always.
5. Vertex form finally justifies why $x = -b/2a$ is the axis of symmetry.

Lecture 10 — From Roots to Vertex: A Unified View

The Vertex as the Midpoint of the Roots, and a Final Check via $x^2 + 1$

1 Recap: The Quadratic Formula

Quadratic Formula (Recap)

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

Writing $D = b^2 - 4ac$, the two roots are

$$x_1 = -\frac{b}{2a} + \frac{\sqrt{D}}{2a}, \quad x_2 = -\frac{b}{2a} - \frac{\sqrt{D}}{2a}.$$

If $D < 0$, no real solution; if $D = 0$, the roots coincide.

2 The Vertex Is the Midpoint of the Roots

Motivating Observation

The two roots differ from $-b/2a$ by $+\sqrt{D}/2a$ and $-\sqrt{D}/2a$ — equal amounts in opposite directions, identifying a midpoint.

Averaging:

$$\frac{x_1 + x_2}{2} = \frac{1}{2} \left[\left(-\frac{b}{2a} + \frac{\sqrt{D}}{2a} \right) + \left(-\frac{b}{2a} - \frac{\sqrt{D}}{2a} \right) \right] = -\frac{b}{2a}.$$

Conclusion

When the roots are real, the x -coordinate of the vertex is their midpoint:

$$x_{\text{vertex}} = \frac{x_1 + x_2}{2} = -\frac{b}{2a}.$$

3 Finding the Y -Coordinate of the Vertex

Substituting $x = -b/2a$ into f :

Vertex y -Coordinate

$$f\left(-\frac{b}{2a}\right) = c - \frac{b^2}{4a} = -\frac{b^2 - 4ac}{4a} = -\frac{D}{4a}.$$

Coordinates of the Vertex

$$\text{Vertex} = \left(-\frac{b}{2a}, -\frac{D}{4a}\right), \quad D = b^2 - 4ac.$$

This single formula packages the axis of symmetry (x -coordinate) and the min/max value (y -coordinate), purely in terms of a, b, c .

4 When There Are No Real Roots: The Midpoint Method Breaks Down

Example: $f(x) = x^2 + 1$, so $a = 1, b = 0, c = 1$:

$$D = b^2 - 4ac = 0 - 4 = -4 < 0.$$

No Roots to Average

Since $D < 0$, there are no real roots — no x_1, x_2 pair to take a midpoint of. The midpoint reasoning does not apply here.

The Vertex Formula Still Works

The coordinate formula does not require real roots:

$$\text{Vertex} = \left(-\frac{b}{2a}, -\frac{D}{4a}\right) = \left(0, -\frac{-4}{4}\right) = (0, 1).$$

Conclusion

The vertex of $f(x) = x^2 + 1$ is $(0, 1)$, above the x -axis; since $a = 1 > 0$ the parabola opens upward and never returns to the x -axis — confirming $x^2 + 1 = 0$ has no real root. Consequently $x^2 + 1$ can never be written in intercept form $a(x - p)(x - q)$ with real p, q .

5 Summary — Lecture 10

Concept	Result
Roots (from quadratic formula)	$x_1, x_2 = -b/2a \pm \sqrt{D}/2a$
Vertex x -coordinate	Midpoint of x_1, x_2 (when real) $= -b/2a$
Vertex y -coordinate	$f(-b/2a) = -D/4a$
Vertex (general formula)	$\left(-\frac{b}{2a}, -\frac{D}{4a}\right)$
Example $f(x) = x^2 + 1$	$D = -4$; vertex $(0, 1)$; no real roots

6 Key Takeaways — Lecture 10

1. The two roots $-b/2a \pm \sqrt{D}/2a$ are symmetric about $-b/2a$; averaging gives a direct derivation of the axis of symmetry.
2. The vertex y -coordinate is $f(-b/2a) = -D/4a$, so the full vertex is $(-b/2a, -D/4a)$.
3. This formula depends only on a, b, c and remains valid even with no real roots.
4. For $f(x) = x^2 + 1$, $D = -4$, yet the vertex formula still gives $(0, 1)$.
5. A quadratic with $D < 0$ can never be written in intercept form with real p, q .

References & Acknowledgements

These notes are based on the video lectures from the Mathematics I course offered at IIT Madras BS Degree Programme.

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Note: These notes may contain some errors. If you spot any, please report them via Discourse so they can be corrected.